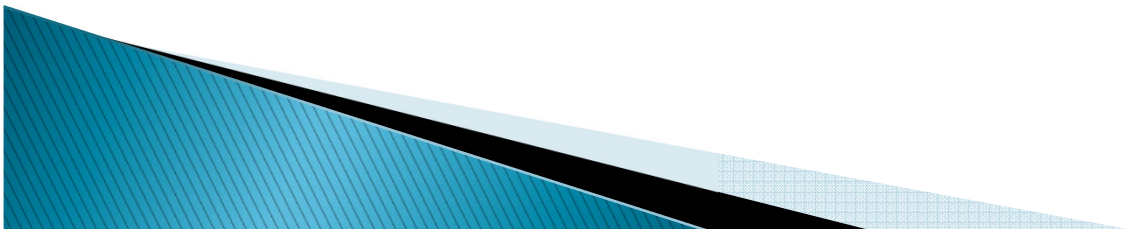


# Example

The figure shows a circular path taken by a particle. If the instantaneous velocity of the particle is  $\vec{v} = (2 \text{ m/s})\hat{i} - (2 \text{ m/s})\hat{j}$  through which quadrant is the particle moving at that instant if it is travelling (a) clockwise and (b) counter-clockwise around the circle? For both cases, draw on the figure.



# Examples

- ▶ Here are four descriptions of the position (in meters) of a puck as it moves in an  $xy$  plane:

1.  $x = -3t^2 + 4t - 2$  and  $y = 6t^2 - 4t$

2.  $x = -3t^3 - 4t$  and  $y = -5t^2 + 6$

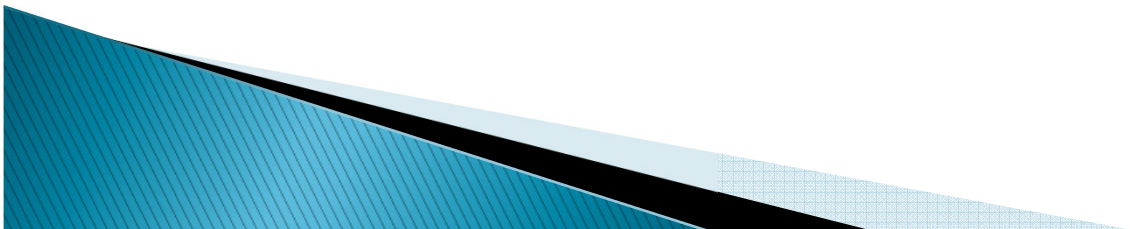
3.  $\vec{r} = 2t^2\hat{i} - (4t + 3)\hat{j}$

4.  $\vec{r} = (4t^3 - 2t)\hat{i} + 3\hat{j}$

Are the  $x$  and  $y$  acceleration components constant? Is the acceleration constant?

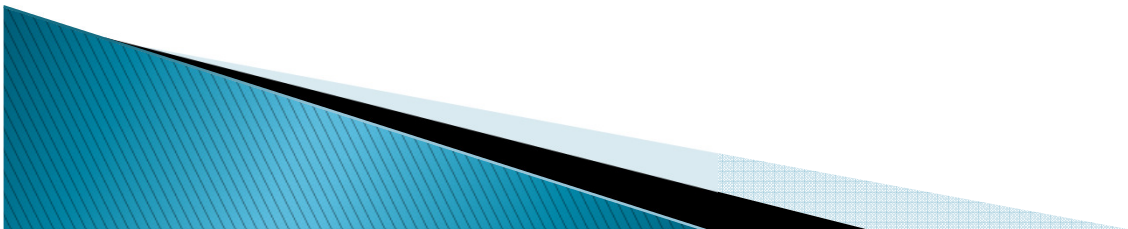
# Example

A particle with velocity  $\vec{v} = -2.0\hat{i} + 4.0\hat{j}$  (in meters per second) at  $t = 0$  undergoes a constant acceleration  $\vec{a}$  of magnitude  $a = 3.0 \text{ m/s}^2$  at an angle  $\theta = 130^\circ$  from the positive direction of the  $x$  axis. What is the particle's velocity  $\vec{v}$  at  $t = 5 \text{ s}$ ?



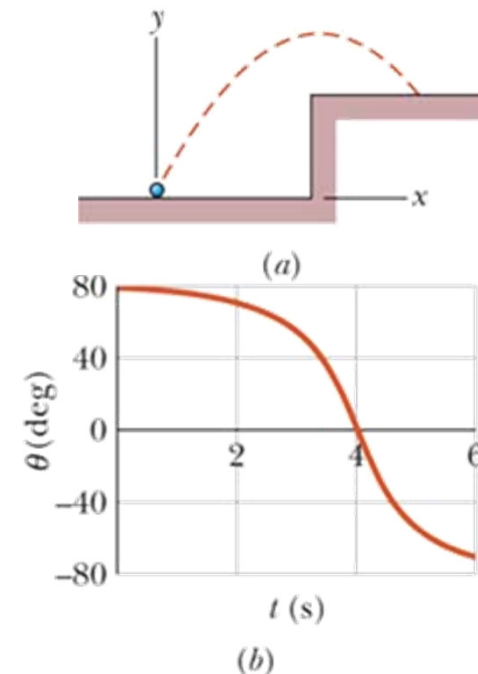
# Problem

- ▶ A rifle with a muzzle velocity  $v_0 = 450$  m/s shoots a bullet at a target 50 m away. How high above the target must the barrel be pointed? Assume the muzzle and the target are at the same height.



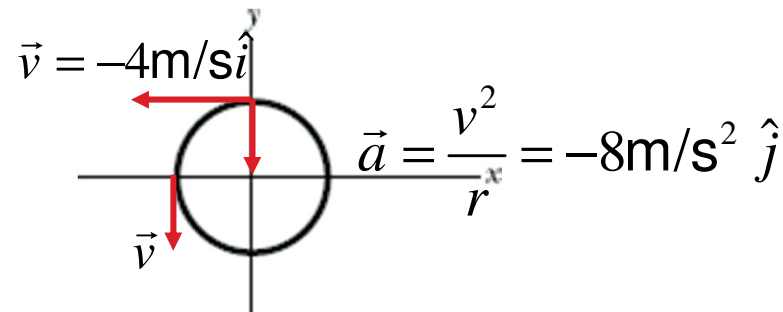
# Problem

- ▶ At time  $t = 0$  a golf ball is shot from ground level into the air, as shown in the fig. The angle  $\theta$  between the ball's direction of travel and the positive  $x$  axis is given as a function of time  $t$ . The ball lands at  $t = 6.00$  s.
  - What is the magnitude  $v_0$  of the ball's launch velocity?
  - At what height  $(y - y_0)$  above the launch level does the ball land?
  - What is the ball's direction of travel just before it lands?



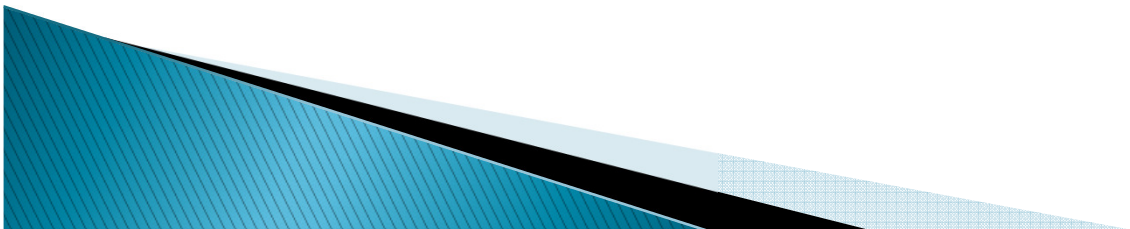
# Example

- ▶ An object moves at constant speed along a circular path in a horizontal  $xy$  plane, with the centre at the origin. When the object is at  $x = -2\text{m}$ , its velocity is  $-(4\text{m/s})\hat{j}$ . Give the object's (a) velocity and (b) acceleration at  $y = 2\text{ m}$ .



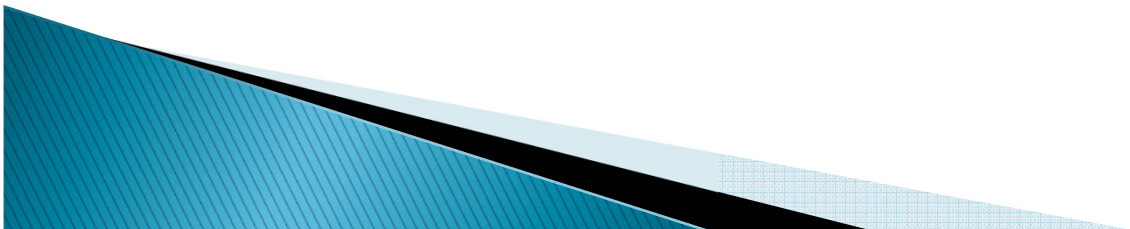
# Problem

- ▶ A satellite is at an altitude  $h = 200$  km above the earth ( $g = 9.2 \text{ m.s}^2$ ). Determine the satellite's velocity and its period.  
The earth's radius is  $6.37 \times 10^6 \text{ m}$ .



# Problem

- ▶ Suppose that Bob's velocity relative to Alex is a constant  $v_{BA} = 52 \text{ km/h}$  and car  $P$  is moving in the negative direction of the  $x$  axis. (a) If Alex measures a constant  $v_{PA} = -78 \text{ km/h}$  for car  $P$ , what velocity  $v_{PB}$  will Bob measure? (b) If car  $P$  brakes to stop relative to Alex in time  $t = 10 \text{ s}$  at constant acceleration, what is its acceleration  $a_{PA}$  relative to Alex? (c) What is the acceleration  $a_{PB}$  of car  $P$  relative to Bob during braking?



# Problem

A plane moves due E while the pilot points the plane somewhat S of E, toward a steady wind that blows to the NE. The plane has velocity  $\vec{v}_{PW}$  relative to the wind, with an airspeed, relative to the wind of 215 km/h directed at angle  $\theta$  S of E. The wind has a velocity  $\vec{v}_{WG}$  relative to the ground with a speed 65 km/h, directed  $20.0^\circ$  E of N. What is the magnitude of the velocity  $\vec{v}_{PG}$  of the plane relative to the ground, and what is  $\theta$ .

